LITTLE STREAM DETECTOR

LSD 3.0 Technical Documentation

Property	Value
Release	3.0
Edition	VPx completion and supported-route cleanup
Runtime	Windows PowerShell 5.1 + .NET Framework
Architecture	Single PowerShell file with embedded C#
External multimedia tools	None
Temporary media files	None
Documentation date	2 September 2026

Current scope. VP8 is supported through Matroska/WebM. VP9 is supported through Matroska/WebM and standard non-fragmented MP4/M4V/MOV. VP8 routing through MP4 `vp08` and VPx routing through AVI are intentionally not enabled.

Known major limit. Fragmented MP4 based on `moof/traf/tfhd/tfdt/trun` is not indexed in LSD 3.0.

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Reading guide

Sections 1-3 describe the stable architecture. Sections 4-7 describe container and codec modules. Sections 8-10 describe measurement, presentation, and validation. Sections 11-12 define the support boundary and regression obligations.

1. Scope and Design Goals

Little Stream Detector 3.0 is a native Windows media inspection application. PowerShell provides orchestration and the WinForms interface, while embedded C# performs binary parsing, canonical sample construction, frame-header traversal, defensive validation, and numerical accumulation.

1.1 Primary objectives

- Analyze supported containers and codecs without launching FFmpeg, MediaInfo, or another multimedia executable.
- Keep container addressing separate from codec interpretation.
- Read only bounded canonical samples that have passed file-range validation.
- Report unavailable information as N/A instead of estimating or inventing values.
- Keep the user interface responsive through staged metadata preparation and background analysis.
- Support one Current A result and one immutable Reference B result for visual comparison.

1.2 Supported families

Area	Supported families
Containers	Matroska, WebM, AVI 1.0, OpenDML AVI 2.0, MP4, M4V, MOV
Video	H.264/AVC, H.265/HEVC, AV1, VP8, VP9, MPEG-4 Part 2/XviD
Audio	AAC-LC, HE-AAC, HE-AACv2, MP3, PCM, AC-3, E-AC-3
Outputs	Summary A/B, Streams, JSON, Log, bitrate profile, quantizer distribution

2. Architecture and Processing Pipeline

LSD is a modular monolith. Container readers and codec analyzers are separate modules connected through a shared canonical model.

```
User / GUI -> native container detection -> track and sample indexing -> canonical media model -> codec
analysis -> completed result snapshot -> reports and A/B graphs
```

Layer	Responsibility	Representative components
Presentation	Window, reports, graphs, drag-and-drop, active-tab copy	PowerShell WinForms
Coordination	Jobs, cancellation, progress, result promotion	Start-Meta, Start-Scan, Finish-Scan
Container parsing	Native tracks, timing, offsets, sample sizes	NativeMediaProbe, NativeMp4Probe, NativeAviProbe
Canonical adaptation	Normalize tracks and bounded samples	MatroskaToLsdAdapter, Mp4ToLsdAdapter, AviToLsdAdapter
Codec analysis	Configuration and frame-level syntax	AVC, HEVC, AV1, VP8, VP9, MPEG-4 Visual
Comparison	Freeze B and render compatible A/B series	ReferenceResult, ReferenceOnlyView

2.1 Processing lifecycle

Phase	Action	Output
1	Select or drop a file	Absolute source path
2	Detect the container	NativeMediaInfo
3	Parse tracks and sample indexes	Container probe result
4	Create bounded canonical samples	LsdMediaModel
5	Traverse codec syntax	Frame and audio diagnostics
6	Finalize reports and histograms	Current A snapshot
7	Optionally promote A to B	Reference B snapshot

3. Canonical Media Model

The canonical layer allows the same codec parser to consume samples originating in different supported containers. A codec parser does not resolve MP4 boxes, Matroska blocks, or AVI chunks.

Object	Core purpose and fields
LsdMediaModel	Source path, container type, file length, duration, timecode scale, tracks
LsdTrackModel	Track ID/type, codec ID/family, codec private bytes, payload format, samples
LsdSampleModel	File offset, length, DTS, PTS, duration, key/invisible/discardable flags
LsdCodecPacket	Container-neutral packet passed to a codec-aware analyzer

3.1 Boundary contract

- FileOffset must be non-negative.
- Length must be positive and compatible with the in-memory parser contract.
- FileOffset + Length must remain within the source file.
- PayloadFormat must match the selected codec analyzer.

3.2 Comparison snapshot

Reference B stores only completed presentation data: file identity, summary text, bitrate bins, quantizer mode, and quantizer counts. It does not keep a second active parser or source stream.

4. Container Support

4.1 Matroska and WebM

The EBML reader discovers native tracks, codec-private data, timestamps, and bounded block payloads. The VPx routes are explicit: V_VP8 selects VP8 and V_VP9 selects VP9.

4.2 Standard MP4, M4V, and MOV

The ISO Base Media reader supports non-fragmented files whose conventional sample tables are stored in `moov`. Supported tables include `stsd`, `stsz`, `stsc`, `stco/co64`, `stts`, `ctts`, and `stss`.

- VP9 sample entry vp09 is mapped to V_VP9 and the shared VP9 parser.
- VP8 sample entry vp08 is intentionally not routed or advertised as supported.
- MOV uses the same standard ISO Base Media sample-table path as MP4/M4V.

4.3 AVI 1.0 and OpenDML AVI 2.0

AVI is supported for the codec combinations listed in the support matrix. VP8 and VP9 are not routed through AVI.

4.4 Fragmented MP4

Fragmented MP4 requires a different sample-indexing path based on `moof`, `traf`, `tfhd`, `tfdt`, and `trun`. LSD 3.0 does not implement that indexer.

5. Video Analysis Modules

Family	Native analysis
H.264 / AVC	avcC and Annex B; SPS/PPS/VUI; I/P/B; SliceQPY; CABAC/CAVLC
H.265 / HEVC	hvcC; VPS/SPS/PPS; NAL class; first-slice I/P/B; SliceQPY; VUI
AV1	av1C; OBU traversal; hidden/show-existing events; Base Q Index
VP8	Frame tag; key/inter; show_frame; boolean-coded first partition; Base Q Index
VP9	Superframe index; uncompressed header; reference slots; shown/hidden; KEY/INTER; Base Q Index
MPEG-4 Part 2 / XviD	VOL/VOP; I/P/B/S-VOP and non-coded VOPs; packed samples; VOP quantizer

5.1 Shared output contract

Each analyzer returns frame classification, completeness diagnostics, rejected-unit information, and an optional codec-specific quantizer histogram. A histogram is shown only when the corresponding parser reports complete results.

5.2 Color metadata

Codec-derived color metadata is supported where the codec parser exposes it. MP4 colr/nclx and Matroska Colour are not currently used as fallback sources when codec-level signaling is absent.

6. VPx Analysis: VP8 and VP9

VP8 and VP9 share container-neutral routing, diagnostics, histogram calculation, GUI presentation, and A/B comparison. The bitstream parsers remain separate because the frame syntax is different.

Feature	VP8	VP9
Supported containers	Matroska/WebM	Matroska/WebM; standard MP4/M4V/MOV via vp09
Frame prefix	3-byte frame tag	Uncompressed frame header
Frame classes	KEY / INTER	KEY / INTER
Display signaling	show_frame	show_frame / show_existing_frame
Multi-frame sample handling	Not required by current route	VP9 superframe index
Frame-level quantizer	Base Q Index, 0-127	Base Q Index, 0-255
Parser core	VP8 boolean decoder	Bounded MSB-first bit reader

6.1 VP8 parser

- Validates the 3-byte frame tag and first-partition size.
- Validates key-frame start code 9D 01 2A and frame dimensions.
- Traverses the boolean-coded segmentation, loop-filter, and quantization syntax.
- Classifies KEY frames as I and INTER frames as P.
- Emits one frame-level Base Q Index from each successfully parsed coded frame.
- Reports parsed, rejected, shown, hidden, segmented, key, and inter counts.

6.2 VP9 parser

- Validates frame marker, profile signaling, sync code, and reference-slot requirements.
- Splits valid superframes and parses internal frames in coded order.
- Separates shown events, hidden coded frames, and show-existing events.
- Maintains reference-slot geometry for frame_size_from_refs.
- Emits Base Q Index only for newly coded frames.

6.3 Quantizer semantics

VP8 and VP9 histograms represent a frame-level base index, not a reconstructed per-block distribution after segmentation. Higher index values generally indicate stronger quantization, but the scale is not a direct linear quality score.

7. Audio Analysis

Family	Validation and reporting
AAC	AudioSpecificConfig, core/output rate, channels, SBR, Parametric Stereo, profile
MP3	Version/layer, bitrate and sample-rate index, padding, frame size, decoded samples
AC-3 / E-AC-3	Syncword, bitstream ID, sample rate, frame size, blocks, channels, LFE
PCM	Block alignment, payload divisibility, decoded duration, expected/measured bitrate

Audio analysis is independent of the active video codec. Adding VP8 or VP9 does not introduce a second audio path and does not change Reference B semantics.

8. Bitrate and Quantizer Analysis

8.1 Bitrate profile

Every canonical video sample contributes its byte length to presentation-time bins. Hidden internal VP9 frames remain in their containing sample and do not create synthetic timing or bitrate bins.

Codec	Native histogram value	Unit
AVC	SliceQPY	Frame-level slice result
HEVC	SliceQPY	Frame-level slice result
AV1	Base Q Index	New coded frame
VP8	Base Q Index	Parsed coded frame
VP9	Base Q Index	New internal coded frame
MPEG-4 Part 2	VOP quantizer	Coded VOP

8.2 Histogram display

- Current A is blue and Reference B is gold.
- Overlays require identical QpMode values.
- VP8 uses the title VP8 Base Q Index distribution and a valid domain of 0-127.
- VP9 uses the title VP9 Base Q Index distribution and a valid domain of 0-255.
- Contextual +/-1 display padding is clamped to a valid codec domain.

9. GUI and A/B Comparison

View	Content
Summary A	Readable current report, including codec-specific diagnostics
Summary B	Frozen reference report while B exists
Streams	Structured native track information
JSON	Serializable current metadata and streams
Log	Lifecycle, validation, and failure messages
Bitrate profile	Blue A and gold B series
Quantizer panel	Mode-aware A/B statistics, bars, and percentages

9.1 State model

```
EMPTY -> open file -> A blue -> Set Ref B -> B gold -> open another file -> B gold + A blue -> Remove Ref B -> A blue, or EMPTY if B was the only result.
```

- Exactly one Current A and one Reference B are supported.
- B is immutable until removed or replaced by explicit promotion.
- B is never silently converted back into A.
- Copy follows the active report tab.

10. Validation and Defensive Parsing

10.1 General rules

- Every sample is checked against file boundaries before payload access.
- Parser readers are bounded to a sample, partition, NAL unit, OBU, or internal VP9 frame.
- Malformed syntax increments a rejected counter and preserves the first failure text.
- No guessed quantizer value is substituted after a parse failure.
- A quantizer graph is enabled only after the relevant completeness checks pass.

10.2 VP8 invariants

- Samples = Parsed + Rejected.
- Parsed = KEY_FRAME + INTER_FRAME.
- Shown + Hidden = Parsed.
- Histogram count = Parsed when Rejected = 0.

10.3 VP9 invariants

- InternalFrames = Parsed + ShowExisting.
- Parsed = KEY_FRAME + INTER_FRAME.
- Histogram count = Parsed when Rejected = 0.
- Displayed I + P + B + Other equals the container sample count for validated routes.

10.4 Runtime isolation

- The source is opened read-only with shared read access.
- No external multimedia process is launched.
- No temporary media file is generated during normal analysis.
- The script remains compatible with in-memory EXE hosting.

11. Support Matrix and Known Limitations

Video	Matroska/WebM	MP4/M4V/MOV	AVI
H.264 / AVC	Yes	Yes	Yes, Annex B
H.265 / HEVC	Yes	Yes	No
AV1	Yes	Yes	No
VP8	Yes	No	No
VP9	Yes	Yes, vp09	No
MPEG-4 Part 2 / XviD	Yes	Yes	Yes

11.1 Current limitations

- Fragmented MP4 is not supported. Only conventional moov-based sample tables are indexed.
- VP8 is supported only in Matroska/WebM. MP4 vp08 and AVI VP8 routes are intentionally absent.
- VP9 is supported in Matroska/WebM and standard non-fragmented MP4/MOV via vp09. VP9 in AVI is not supported.
- VP8 and VP9 histograms report a frame-level base index, not a per-block reconstruction after segmentation.
- Container color metadata is not used as fallback when codec-level color signaling is missing.
- LSD does not decode pixels and does not calculate PSNR, SSIM, VMAF, or related image metrics.
- The UI supports one A/B pair, not batch ranking.

12. Regression Matrix and Maintenance Rules

Asset	Expected result
VP8 Matroska/WebM regression sample	300 frames; 2 KEY; 298 INTER; 300 shown; Base Q 4-126; average 64.210000; zero rejected
VP9 10-second Matroska sample	300 samples; 325 internal frames; 25 superframes; 2 KEY; 323 INTER; Base Q 31-179; zero rejected
VP9 standard MP4 sample	300 samples; 300 internal frames; 2 KEY; 298 INTER; Base Q 46-190; zero rejected

12.1 Stable-code rules

- Keep VP8 and VP9 bitstream parsers independent.
- Keep Matroska/WebM V_VP8 routing and MP4 vp09 routing explicit.
- Do not reintroduce MP4 vp08 or AVI VPx routing without a representative regression asset and an explicit support decision.
- Do not change canonical model contracts while adding a codec-specific parser.
- Preserve A/B role semantics and snapshot immutability.
- Preserve N/A reporting and rejected-unit visibility.

12.2 Future candidate

The remaining major container extension is fragmented MP4 sample indexing. It should be implemented as a separate indexer that resolves trex/tfhd/tfdt/trun state into the existing canonical sample model. Codec parsers should remain unchanged.

Appendix A. Key Types and Responsibilities

Type / state	Responsibility
NativeMediaInfo / NativeTrack	Container-native metadata and codec configuration
LsdMediaModel / LsdTrackModel / LsdSampleModel	Container-neutral tracks and bounded samples
NativeMediaProbe / NativeMp4Probe / NativeAviProbe	Container detection, metadata, and sample indexing
NativeMatroskaPacketScan	Shared canonical video analysis job; historical type name
Vp8BoolDecoder / Vp8FrameParser	VP8 first-partition traversal and Base Q extraction
Vp9BitReader / Vp9FrameParser	VP9 superframe and uncompressed-header traversal
CanonicalAudioAnalyzer	Native MP3, Dolby, PCM, and sample accounting
ReferenceResult	Frozen B summary, bitrate bins, QP mode, and histogram
PowerShell coordinator	Jobs, reports, graphs, comparison, and GUI state

Appendix B. Terminology

Term	Definition
Canonical sample	Container-neutral record describing a bounded payload range and timing
VP8 Base Q Index	Frame-level VP8 quantizer index in the domain 0-127
Internal VP9 frame	One coded VP9 frame extracted directly or from a superframe payload
VP9 superframe	One sample containing concatenated coded frames and a trailing size index
Shown frame event	Presentation caused by a newly coded shown frame or show_existing_frame
Hidden coded frame	A newly coded frame with show_frame equal to zero
Base Q Index	Frame-level native quantizer lookup index; 0-127 for VP8 and 0-255 for VP9/AV1
Current A	Most recently analyzed active result, shown in blue
Reference B	Stored comparison result, shown in gold
Complete	All required relevant units parsed without rejection

Release identification. Little Stream Detector 3.0 remains a single PowerShell 5.1 application with embedded C#, no external multimedia tools, and no temporary media files.